

In [1]: `import numpy as np`

```
# Create a numpy array from 1 to 10
arr1 = np.arange(1, 11)

# Reshape it to a 2x5 matrix
reshaped_arr1 = arr1.reshape(2, 5)

print("Exercise 1:")
print("Original array:")
print(arr1)
print("Reshaped array:")
print(reshaped_arr1)
```

Exercise 1:
Original array:
[1 2 3 4 5 6 7 8 9 10]
Reshaped array:
[[1 2 3 4 5]
 [6 7 8 9 10]]

In [2]: `# Create a numpy array from 1 to 20`
`arr2 = np.arange(1, 21)`

```
# Extract elements between the 5th and 15th index
extracted_elements = arr2[4:15]

print("\nExercise 2:")
print("Extracted elements:")
print(extracted_elements)
```

Exercise 2:
Extracted elements:
[5 6 7 8 9 10 11 12 13 14 15]

In [3]: `import pandas as pd`

```
# Create a Pandas series
fruits_data = {'apples': 3, 'bananas': 2, 'oranges': 1}
fruits_series = pd.Series(fruits_data)

# Add a new item to the series
fruits_series['pears'] = 4

print("\nExercise 3:")
print("Pandas series:")
print(fruits_series)
```

Exercise 3:
Pandas series:
apples 3
bananas 2
oranges 1
pears 4
dtype: int64

In [4]: `# Create a DataFrame with 10 rows`

```
data = {
    'name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank', 'Grace', 'Hannah', 'Ivy', 'Jack'],
    'age': [25, 32, 28, 40, 22, 35, 29, 42, 31, 27],
    'gender': ['F', 'M', 'F', 'M', 'F', 'M', 'F', 'F', 'F', 'M']
}

df = pd.DataFrame(data)

print("\nExercise 4:")
print("DataFrame:")
print(df)
```

Exercise 4:
DataFrame:
 name age gender
0 Alice 25 F
1 Bob 32 M
2 Charlie 28 F
3 David 40 M
4 Eve 22 F
5 Frank 35 M
6 Grace 29 F
7 Hannah 42 F
8 Ivy 31 F
9 Jack 27 M

```
In [5]: # Step 1: Add the 'occupation' column
df['occupation'] = ['Programmer', 'Manager', 'Analyst', 'Programmer', 'Manager',
                   'Analyst', 'Programmer', 'Manager', 'Analyst', 'Programmer']
print(df)
```

	name	age	gender	occupation
0	Alice	25	F	Programmer
1	Bob	32	M	Manager
2	Charlie	28	F	Analyst
3	David	40	M	Programmer
4	Eve	22	F	Manager
5	Frank	35	M	Analyst
6	Grace	29	F	Programmer
7	Hannah	42	F	Manager
8	Ivy	31	F	Analyst
9	Jack	27	M	Programmer

```
In [6]: # Select rows where age is greater than or equal to 30
selected_rows = df[df['age'] >= 30]

print("\nExercise 6:")
print("Selected rows:")
print(selected_rows)
```

Exercise 6:

Selected rows:

	name	age	gender	occupation
1	Bob	32	M	Manager
3	David	40	M	Programmer
5	Frank	35	M	Analyst
7	Hannah	42	F	Manager
8	Ivy	31	F	Analyst

```
In [7]: # Convert DataFrame to CSV
df.to_csv('my_dataframe.csv', index=False)

# Read the CSV file
read_df = pd.read_csv('my_dataframe.csv')

print("\nExercise 7:")
print("Read DataFrame from CSV:")
print(read_df)
```

Exercise 7:

Read DataFrame from CSV:

	name	age	gender	occupation
0	Alice	25	F	Programmer
1	Bob	32	M	Manager
2	Charlie	28	F	Analyst
3	David	40	M	Programmer
4	Eve	22	F	Manager
5	Frank	35	M	Analyst
6	Grace	29	F	Programmer
7	Hannah	42	F	Manager
8	Ivy	31	F	Analyst
9	Jack	27	M	Programmer

```
In [ ]:
```

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